Welcome to 18.01

Welcome to 18.01. Today we start "Unit One"; the topic of the unit is differentiation. We'll start by reviewing what's in store in the next couple of weeks.

The topic of this lecture is "what is a derivative?" We're going to look at this question from several different points of view, and the first one is the geometric interpretation. We'll also discuss a physical interpretation.

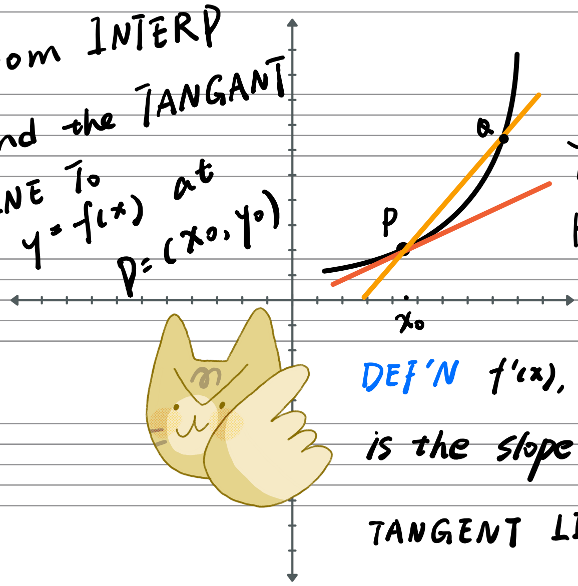
Later we'll learn what makes calculus so fundamental in science and engineering. Derivatives are important in all measurements – in science, in engineering, in economics, in political science, in polling, in lots of commercial applications, in just about everything.

In this unit we'll also learn how to differentiate any function you know. That's a tall order, but by the end of the unit you will know how to take derivatives of functions like $f(x) = e^{x \cdot \arctan(x)}$.

Let's begin.

Geom INTERP

Find the TANGENT LINE To $y = f(x)$ at $P = (x_0, y_0)$

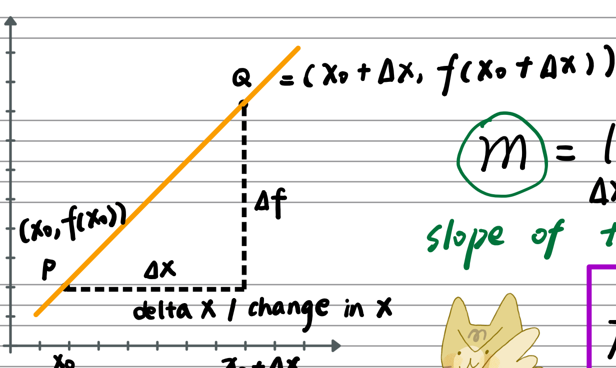


$y - y_0 = m(x - x_0)$ slope $m = f'(x)$

point $y_0 = f(x_0)$

DEF'N $f'(x)$, the derivative of f at x_0, is the slope of the tangent line to $y = f(x)$ at P.

TANGENT LINE = LIMIT of Secant LINES PQ as $Q \to P$



$$m = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x}$$

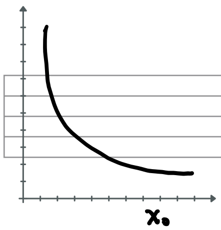
slope of secant PQ

slope of tangent

$$\boxed{\underset{(=m)}{f'(x_0)} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}}$$

difference quotatio

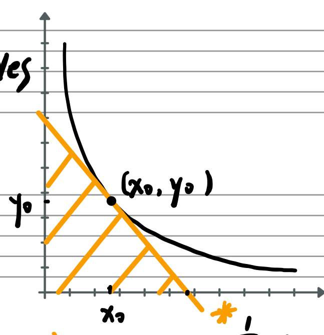
Example 1 $f(x) = \frac{1}{x}$ $\boxed{f'(x_0) = -\frac{1}{x_0^2}}$ <0 negative slope, $x_0 \to \infty$, less & less sleep.



$$\frac{\Delta f}{\Delta x} = \frac{\frac{1}{x_0+\Delta x} - \frac{1}{x_0}}{\Delta x} = \frac{1}{\Delta x}\left(\frac{x_0-(x_0+\Delta x)}{(x_0+\Delta x)x_0}\right) = \frac{1}{\Delta x}\left(\frac{-\Delta x}{(x_0+\Delta x)x_0}\right)$$

$$= \frac{-1}{(x_0+\Delta x)x_0} \xrightarrow[\Delta x \to 0]{} \boxed{-\frac{1}{x_0^2}}$$

Find Areas of Triangles enclosed by Axes and TANGENT to $y = \frac{1}{x}$



(*) $y - y_0 = -\frac{1}{x_0^2}(x - x_0)$

Area of Δ:

$\frac{1}{2}(2x_0)(2y_0) = 2x_0y_0 = 2$

Find x-intercept
$(y=0)$

$0 - \frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0) = -\frac{x}{x_0^2} + \frac{1}{x_0}$

$\Rightarrow \frac{x}{x_0^2} = \frac{2}{x_0} \Rightarrow \boxed{x = 2x_0}$

Shortcut to y-Intercept
use symmetry $y = 2y_0$

$(x,y) \to (y,x)$

Symmetry explanation

$y = \frac{1}{x} \iff xy = 1 \iff x = \frac{1}{y}$

Can also get y-intercept by plugging $x=0$ into (*)

More Notations

$$f' = \underbrace{\frac{df}{dx} = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}y}_{\text{Leibniz}}$$

$y = f(x)$, $\Delta y = \Delta f$ Newton's

(omits x_0)

Example 2. $f(x) = x^n$, $n = 1, 2, 3, \cdots$ (x is fixed, Δx moves)

¿ $\frac{d}{dx}x^n = ?$ $\frac{\Delta f}{\Delta x} = \frac{(x+\Delta x)^n - x^n}{\Delta x}$

$\frac{d}{dx}x^n = nx^{n-1}$ Extends to polynomials

$$\frac{d}{dx}(x^3 + 5x^{10}) = 3x^2 + 50x^9$$

(bionomial theorem)

$$(x+\Delta x)^n = (x+\Delta x)\cdots(x+\Delta x)$$

$$= x^n + nx^{n-1}\Delta x + \text{junk } (O(\Delta x)^2) \text{ terms } w/\ (\Delta x)^2, (\Delta x)^3 \ldots$$

big O of delta x squared

$$\frac{\Delta f}{\Delta x} = \frac{1}{\Delta x}\left((x+\Delta x)^n - x^n\right)$$

$$= \frac{1}{\Delta x}\left(x^n + nx^{n-1}\Delta x + O((\Delta x)^2) - x^n\right)$$

$$= \frac{1}{\Delta x}\left(nx^{n-1}\Delta x + O(\Delta x)^2\right).$$

$$= nx^{n-1} + O(\Delta x)$$

$$\xrightarrow[\Delta x \to 0]{} nx^{n-1}$$